SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		Slot: B
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	CSE AI	**Date:**	05-08-2026

Code of Conduct

I P Mohan Babu / 192472254 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P MohanBabu

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

- Develop a Python program to implement a rule-based morphological processing system for the input words "analyzing", "analysis", and "analytical". The program should identify the root word and separate the corresponding prefixes or suffixes using suitable morphological rules, classify each word as an inflectional or derivational form, normalize all related variants to a common representation for indexing, and generate a structured report showing the original word, extracted root, identified affix(es), transformation type, and normalized output to support efficient search engine retrieval.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

- Develop a Python program to create a morphological parser for the input words "disagree", "agreement", and "agreeable". The program should detect and separate prefixes, suffixes, and the base word using rule-based morphological analysis, examine how derivational modifications alter the meaning of the root word, classify each transformation according to its morphological type, and produce a comprehensive output containing the original word, identified prefix, root word, suffix, transformation category, semantic interpretation, and normalized base form for use in sentiment analysis applications.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

- Develop a Python program to implement a morphology-based normalization module for the input words "govern", "government", and "governance". The program should perform morphological decomposition to identify the root word and successive derivational suffixes, determine the derivational level associated with each word, normalize all variants to a common lexical representation, and generate a structured output displaying the original word, root form, detected affix(es), derivational hierarchy, normalized representation, and the final output used for topic modeling and document clustering.

4.A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation.
Derive the morphological structure and normalized root for each input word.

- Develop a Python program to implement a morphological parsing and normalization system for the input words "activate", "activation", and "reactivation". The program should detect and separate prefixes, root words, and suffixes using rule-based morphological parsing, identify the sequence of derivational transformations that produce each word, analyze the effect of each morphological modification on the word class and semantic meaning, and generate a structured report showing the original word, prefix (if present), root word, suffix, derivational sequence, normalized base form, and final parsed representation for use in document classification and semantic indexing.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation.
Determine the morphological breakdown and normalized output for each word.

- Develop a Python program to implement an inflectional morphology-based normalization module for the input words "create", "creates", and "creating". The program should recognize grammatical suffixes corresponding to different inflectional forms, identify the underlying root word by applying appropriate normalization rules, classify each input according to its grammatical feature (e.g., base form, third-person singular, present participle), and produce a structured output containing the original word, identified suffix, grammatical category, extracted root, normalized base form, and the final normalized representation to support efficient search optimization and information retrieval.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Morphological Processing of "analyzing", "analysis", and "analytical"

Aim

Design a rule-based morphological processing system that:

- Identifies root and affixes
- Distinguishes inflectional and derivational forms
- Normalizes all words to a common representation for indexing

Python Program

```
# Rule-Based Morphological Processing

words = ["analyzing", "analysis", "analytical"]

def analyze_word(word):
    root = ""
    affix = ""
    transformation = ""

    if word.endswith("ing"):
        root = word[:-3]
        if root.endswith("z"):
            root = root[:-1] + "ze"
        affix = "-ing"
        transformation = "Inflectional"

    elif word.endswith("sis"):
        root = "analyze"
        affix = "-sis"
        transformation = "Derivational"

    elif word.endswith("ical"):
        root = "analyze"
        affix = "-ical"
        transformation = "Derivational"

    else:
        root = word
        transformation = "Base"

    normalized = "analyze"

    return [word, root, affix, transformation, normalized]

print("{:<15}{:<12}{:<10}{:<15}{:<12}".format(
    "Word","Root","Affix","Type","Normalized"))

for w in words:
    result = analyze_word(w)
    print("{:<15}{:<12}{:<10}{:<15}{:<12}".format(*result))
```

Expected Output

Original Word	Root	Affix	Transformation	Normalized
analyzing	analyze	-ing	Inflectional	analyze
analysis	analyze	-sis	Derivational	analyze
analytical	analyze	-ical	Derivational	analyze

Explanation

- **analyzing** → analyze + ing (Present participle → Inflectional)
- **analysis** → analyze + sis (Noun formation → Derivational)
- **analytical** → analyze + ical (Adjective formation → Derivational)

All words normalize to **analyze** for efficient search indexing.

OUTPUT:

Word	Root	Affix	Type	Normalized
analyzing	analyze	-ing	Inflectional	analyze
analysis	analyze	-sis	Derivational	analyze
analytical	analyze	-ical	Derivational	analyze

Q2. Morphological Parsing of "disagree", "agreement", and "agreeable"

Aim

Detect prefixes, suffixes, root words, derivational transformations, semantic interpretation, and normalized base.

Python Program

```
words = ["disagree", "agreement", "agreeable"]

def parse(word):

    prefix = ""
    suffix = ""
    root = "agree"

    if word.startswith("dis"):
        prefix = "dis-"
        category = "Derivational"
        meaning = "Negation"

    elif word.endswith("ment"):
        suffix = "-ment"
        category = "Derivational"
        meaning = "Action/Result (Noun)"

    elif word.endswith("able"):
        suffix = "-able"
        category = "Derivational"
        meaning = "Capable of"

    else:
        category = "Base"
        meaning = "Agreement"

    normalized = "agree"

    return [word,prefix,root,suffix,category,meaning,normalized]

print("{:<15}{:<8}{:<10}{:<10}{:<15}{:<20}{:<12}".format(
    "Word","Prefix","Root","Suffix","Category","Meaning","Normalized"))

for w in words:
    print("{:<15}{:<8}{:<10}{:<10}{:<15}{:<20}{:<12}".format(*parse(w)))
```

Expected Output

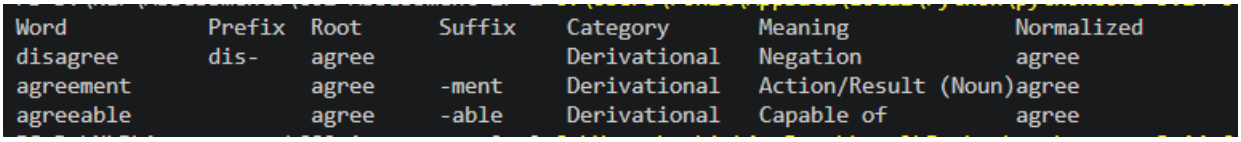
Word	Prefix	Root	Suffix	Category	Semantic Interpretation	Normalized
disagree	dis-	agree	—	Derivational	Negation	agree
agreement	—	agree	-ment	Derivational	Action/Result	agree
agreeable	—	agree	-able	Derivational	Capable of agreement	agree

---

Explanation

- **dis-** changes meaning (negative).
- **-ment** converts verb → noun.
- **-able** converts verb → adjective.
- All normalize to **agree**.

OUTPUT:



```
Word           Prefix  Root      Suffix    Category       Meaning              Normalized
disagree       dis-    agree               Derivational   Negation             agree
agreement              agree     -ment     Derivational   Action/Result (Noun)agree
agreeable              agree     -able     Derivational   Capable of           agree
```

Q3. Morphology-Based Normalization of "govern", "government", and "governance"

Aim

Identify derivational hierarchy and normalize to a common lexical representation.

Python Program

```
words = ["govern","government","governance"]

def normalize(word):

    root = "govern"
    affix = ""
    level = ""

    if word == "govern":
        level = "Level 0"

    elif word.endswith("ment"):
        affix = "-ment"
        level = "Level 1"

    elif word.endswith("ance"):
        affix = "-ance"
        level = "Level 1"

    normalized = "govern"

    return [word,root,affix,level,normalized]

print("{:<15}{:<10}{:<10}{:<15}{:<15}".format(
    "Word","Root","Affix","Hierarchy","Normalized"))

for w in words:
    print("{:<15}{:<10}{:<10}{:<15}{:<15}".format(*normalize(w)))
```

Expected Output

Word	Root	Affix	Hierarchy	Normalized
govern	govern	—	Level 0	govern

Word	Root	Affix	Hierarchy	Normalized
government	govern	-ment	Level 1	govern
governance	govern	-ance	Level 1	govern

Explanation

- **govern** → Verb
- **government** → Noun (institution)
- **governance** → Noun (process/system)

All normalize to **govern**.

OUTPUT:

```
Word          Root      Affix     Hierarchy      Normalized
govern        govern                Level 0        govern
government    govern    -ment       Level 1        govern
governance    govern    -ance       Level 1        govern
```

Q4. Morphological Parsing of "activate", "activation", and "reactivation"

Aim

Identify prefixes, suffixes, derivational sequence, semantic changes, and normalization.

Python Program

```
words = ["activate","activation","reactivation"]

def parse(word):

    prefix = ""
    suffix = ""
    root = "activate"

    if word == "activate":
        sequence = "Base Verb"

    elif word == "activation":
        suffix = "-ion"
        sequence = "activate + ion"

    elif word == "reactivation":
        prefix = "re-"
        suffix = "-ion"
        sequence = "re + activate + ion"

    normalized = "activate"

    return [word,prefix,root,suffix,sequence,normalized]

print("{:<18}{:<8}{:<12}{:<10}{:<25}{:<12}".format(
    "Word","Prefix","Root","Suffix","Sequence","Normalized"))

for w in words:
    print("{:<18}{:<8}{:<12}{:<10}{:<25}{:<12}".format(*parse(w)))
```

Expected Output

Word	Prefix	Root	Suffix	Derivational Sequence	Normalized
activate	—	activate	—	Base Verb	activate
activation	—	activate	-ion	activate → activation	activate
reactivation	re-	activate	-ion	re + activate + ion	activate

Explanation

- **-ion** changes verb → noun.
- **re-** means "again".
- Final normalized form = **activate**.

OUTPUT:

```
Word            Prefix  Root      Suffix   Sequence               Normalized
activate                activate           Base Verb              activate
activation              activate  -ion     activate + ion         activate
reactivation    re-     activate  -ion     re + activate + ion    activate
```

Q5. Inflectional Morphology of "create", "creates", and "creating"

Aim

Recognize grammatical suffixes and normalize all words to the root.

Python Program

```
words = ["create","creates","creating"]

def normalize(word):

    suffix = ""
    category = ""
    root = "create"

    if word == "create":
        category = "Base Form"

    elif word.endswith("s"):
        suffix = "-s"
        category = "Third Person Singular"

    elif word.endswith("ing"):
        suffix = "-ing"
        category = "Present Participle"

    normalized = "create"

    return [word,suffix,category,root,normalized]

print("{:<15}{:<10}{:<25}{:<12}{:<12}".format(
    "Word","Suffix","Category","Root","Normalized"))

for w in words:
    print("{:<15}{:<10}{:<25}{:<12}{:<12}".format(*normalize(w)))
```

Expected Output

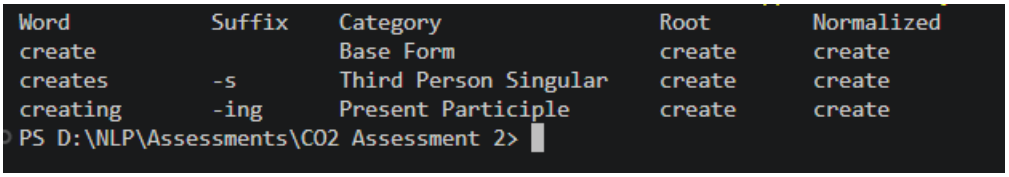
Word	Suffix	Grammatical Category	Root	Normalized
create	—	Base Form	create	create
creates	-s	Third Person Singular	create	create
creating	-ing	Present Participle	create	create

Explanation

- **create** → Base verb.
- **creates** → Third-person singular (Inflectional).
- **creating** → Present participle/continuous aspect (Inflectional).

- Meaning remains unchanged, so all normalize to **create**.

OUTPUT:



Conclusion

These rule-based morphological analyzers:

- Extract **roots, prefixes, and suffixes**.
- Distinguish **inflectional** vs. **derivational** morphology.
- Normalize related word forms to a common base (lemma).
- Produce structured outputs suitable for **search engines, sentiment analysis, topic modeling, document classification, and information retrieval**.